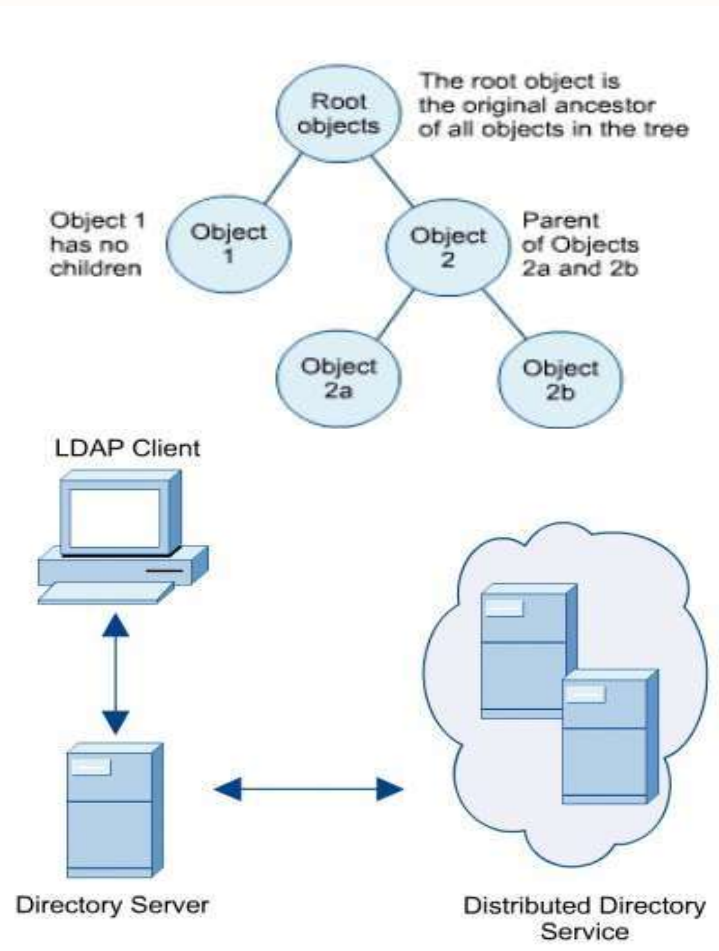


Network Directory Services

6CS029 Advanced Networks



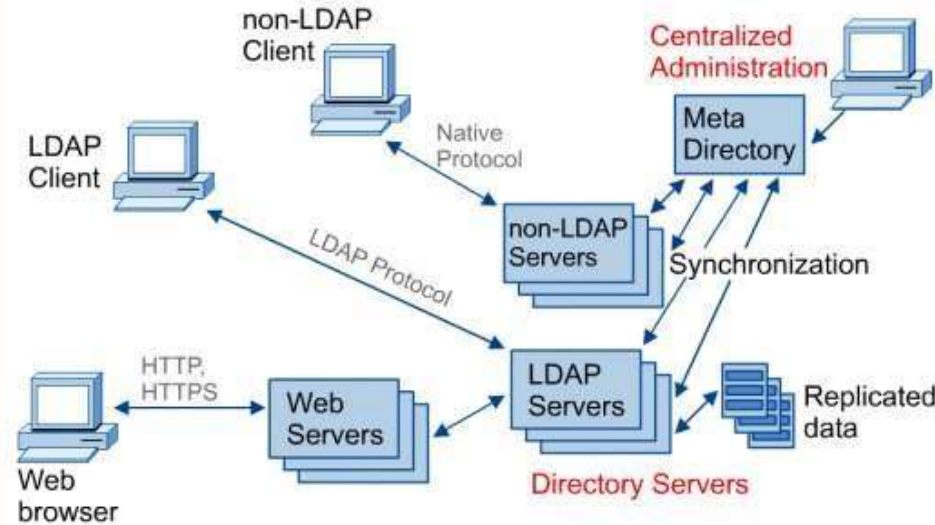
What is a Directory Service?



- A directory service provides system administrators with centralized control of all users and resources across the entire network.
- They provide the ability to organize information and help simplify the management of the network by providing a standard interface for common system administration tasks. Shared resources are published to the directory
- Users can locate and access them without ever knowing on which machine the resources physically reside.
- The files, directories, and shares that users access from a single point can be distributed across multiple servers and locations using distributed directory and replication services.



Directory Service Standards

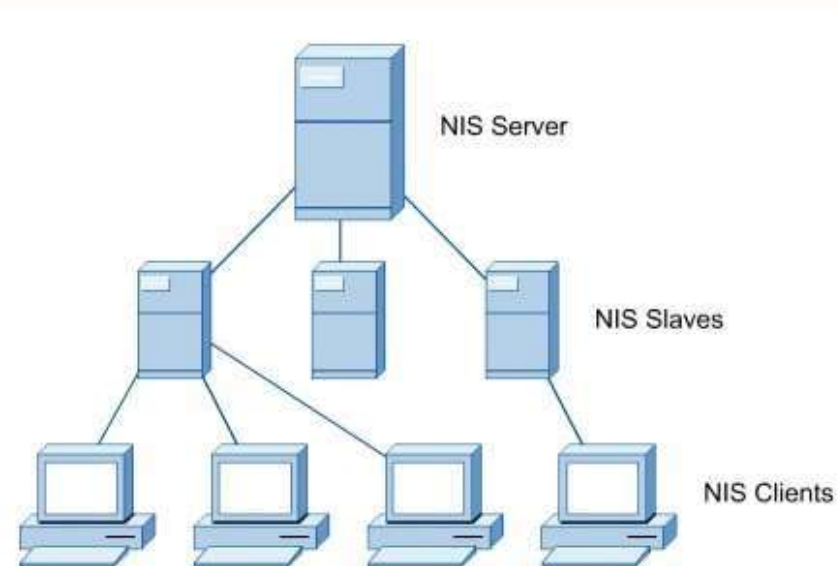


- To operate within a network, different directory services need to have a common method of naming and referencing objects.
- X.500 defines the Electronic Directory Service (EDS) standards.
- Lightweight Directory Access Protocol (LDAP) defines an Internet-based access method



Network Information Service (NIS)

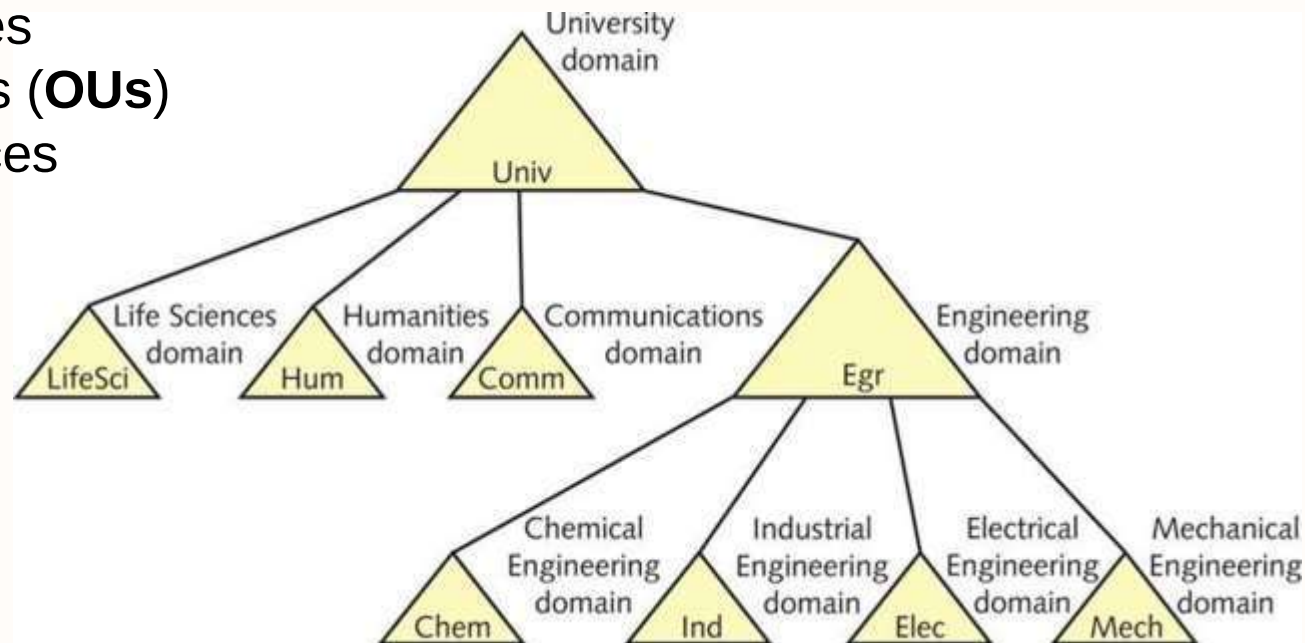
- The Directory Service used in most Unix/Linux systems is called the Network Information Service (NIS).
- The network consists of the NIS server, slaves, and clients.
- The NIS Servers is where the NIS database is created and maintained.
- The NIS databases are copied to all the NIS slave servers.





Windows Active Directory

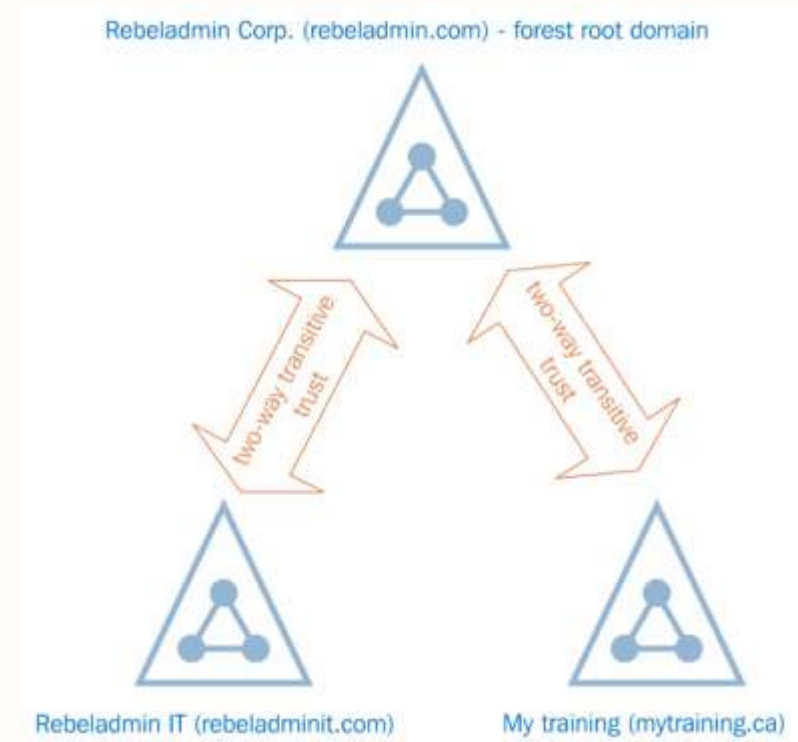
- The logical structure of the Active Directory is based on units called **Domains**.
- Windows networks can have multiple domains, organized into **domain trees**.
- These trees can be joined to other trees to form **forests**.
- Active Directory uses **Organizational Units (OUs)** to organize resources within domains





Forests

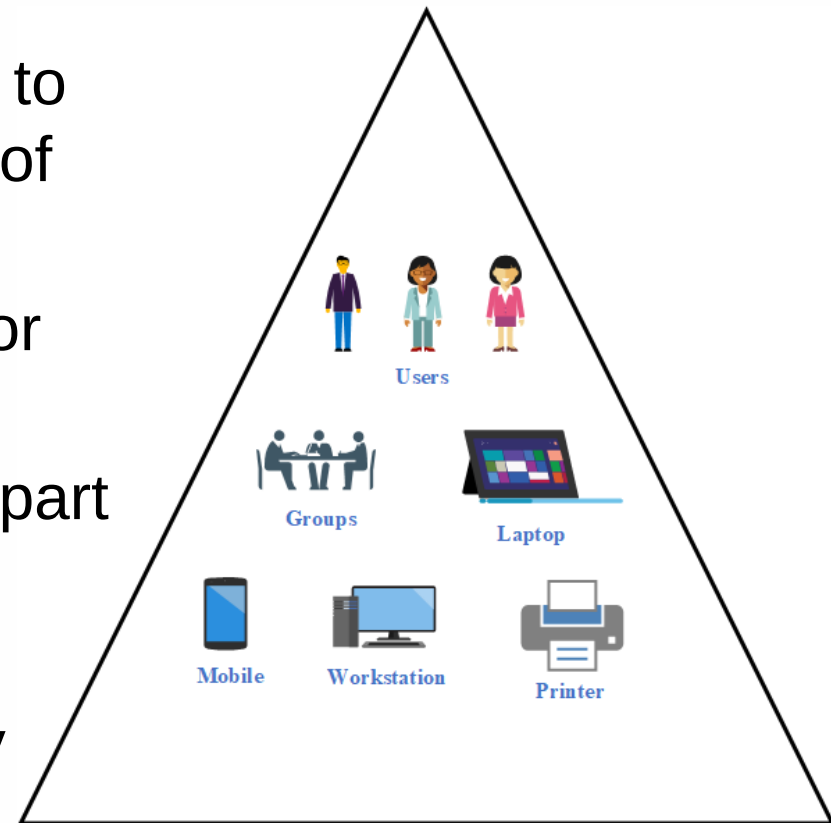
- Represents a complete Active Directory instance
- It is made of one or more domains and domain trees.
- A domain tree contains its own root domain, but forests can contain multiple root domains
- Two-way transitive trust is a logical link between domains





Domains

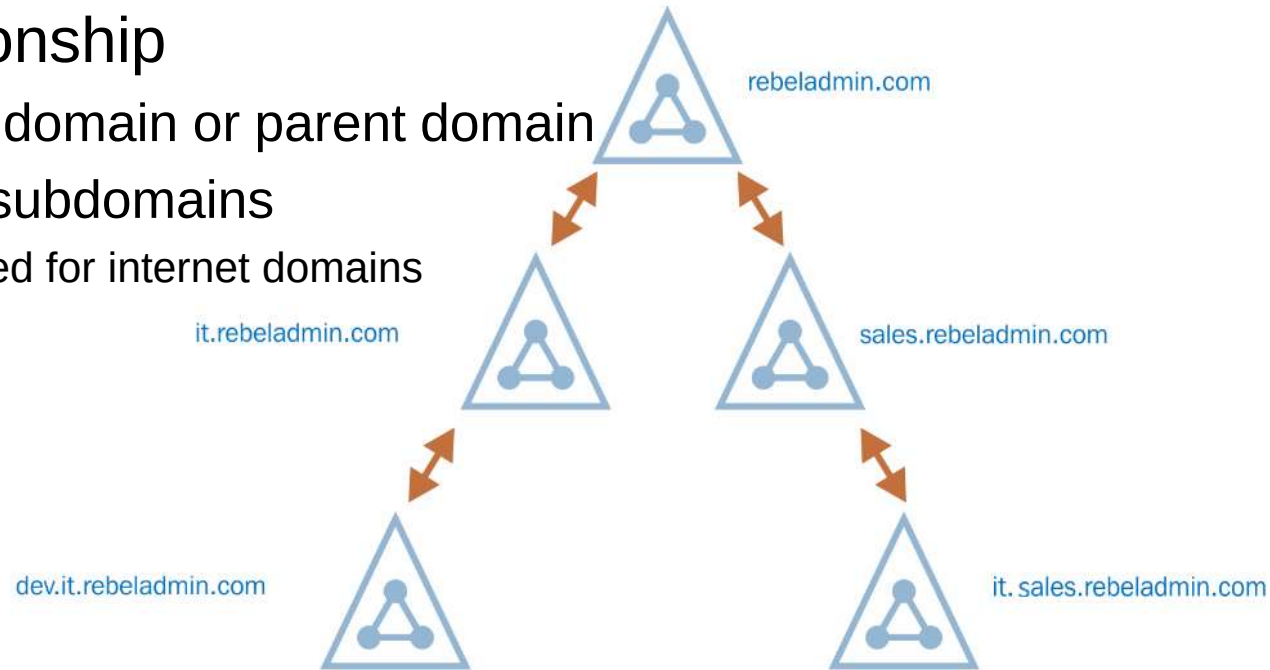
- Contains the logical components to achieve the administrative goals of an organization
- Provides the security boundary for the objects inside it
- All the objects in the domain are part of a common database
- Objects in the domain are also controlled by the defined security rules





Domain Trees

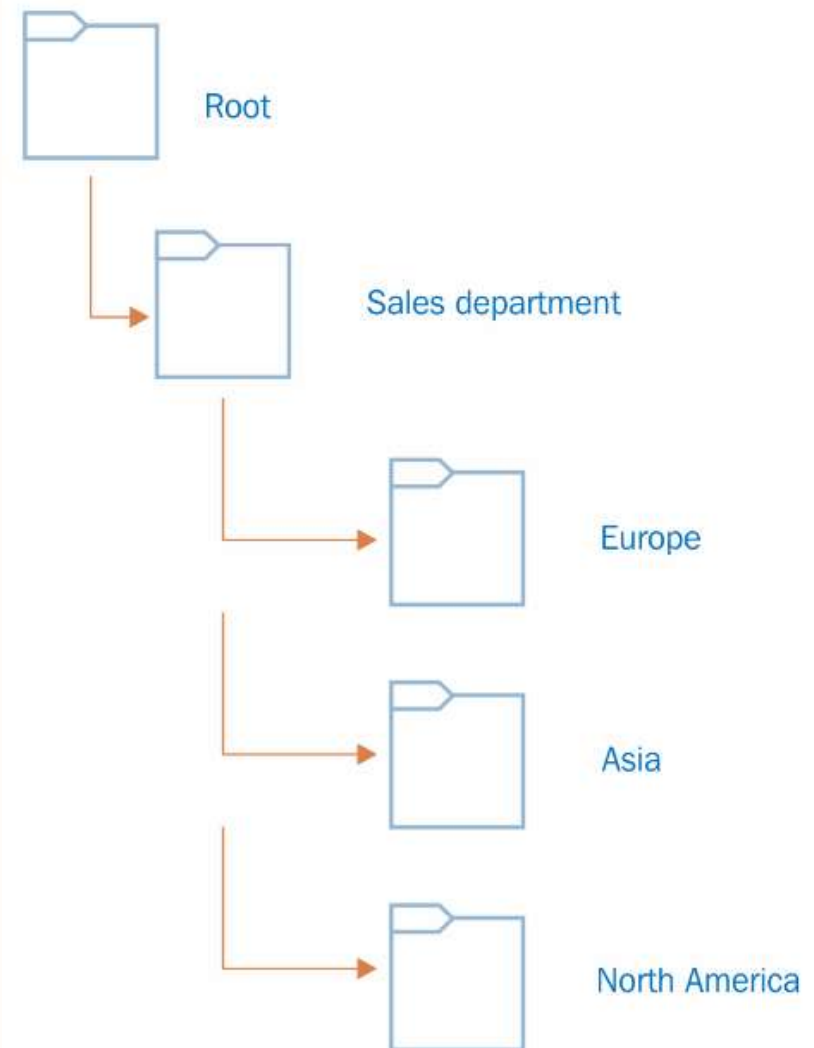
- Collection of domains that reflects the organization's structure
- Domains inside the domain tree have a parent-child relationship
 - First domain- root domain or parent domain
 - Child domains or subdomains
 - sub-URL is required for internet domains





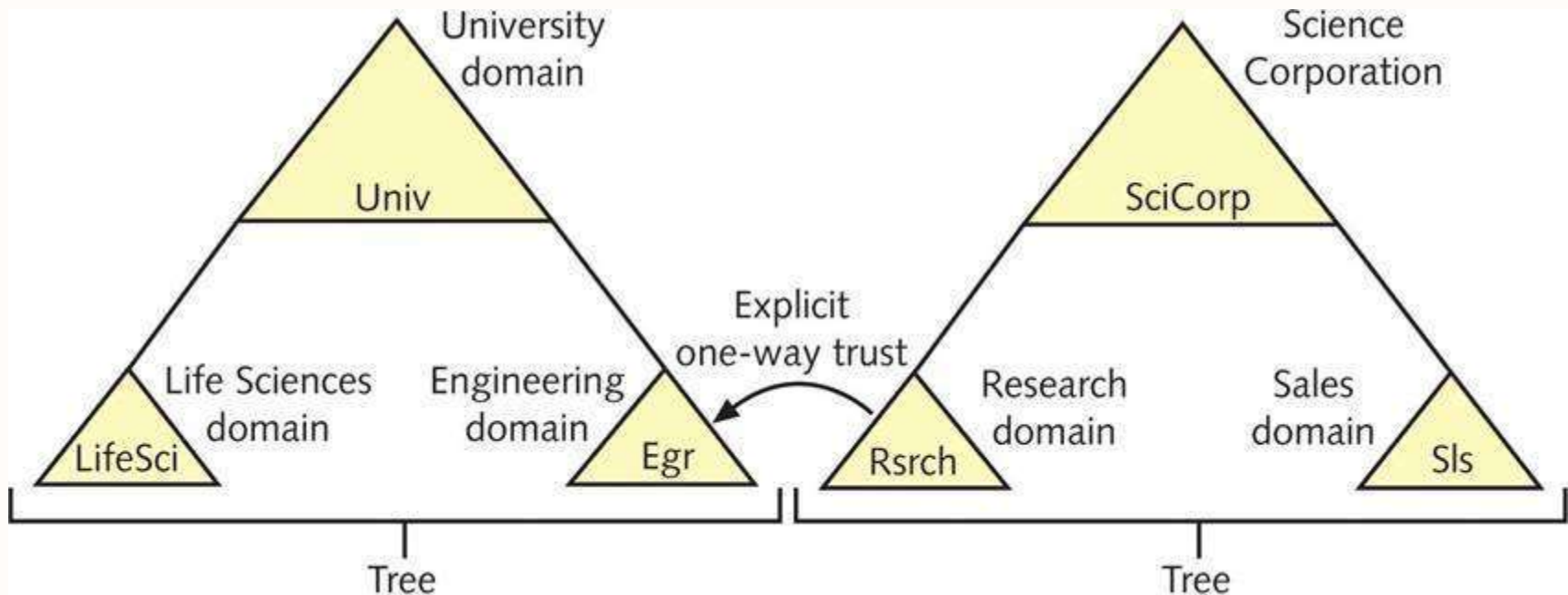
OUs (Organizational Units)

- OUs group objects into a logical hierarchy
- Delegate administrative control over the objects within an OU by assigning specific permissions to users and groups





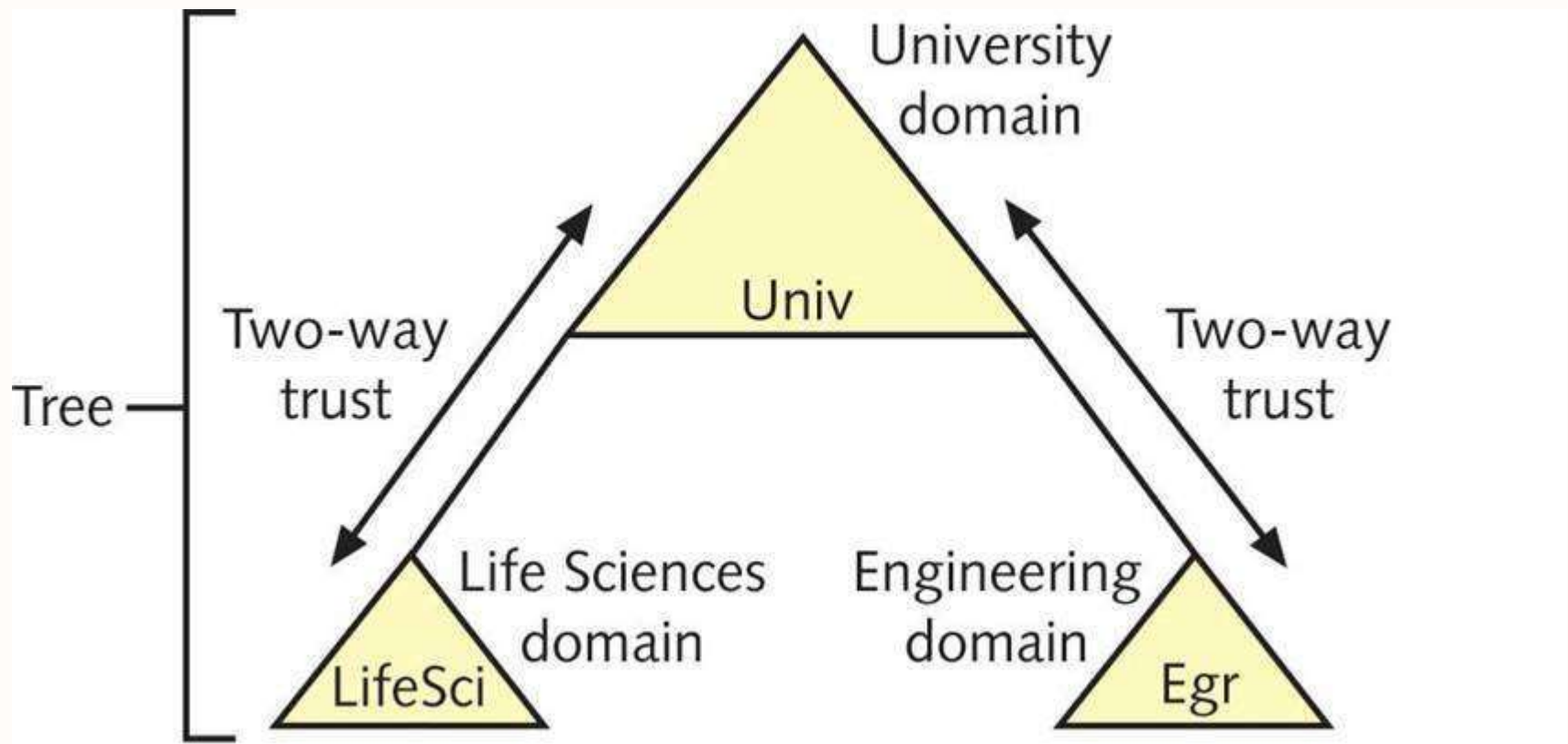
One Way Trust Relationships



Explicit one-way trust between domains in different trees



Trust Relationships



Two-way trusts between domains in a tree

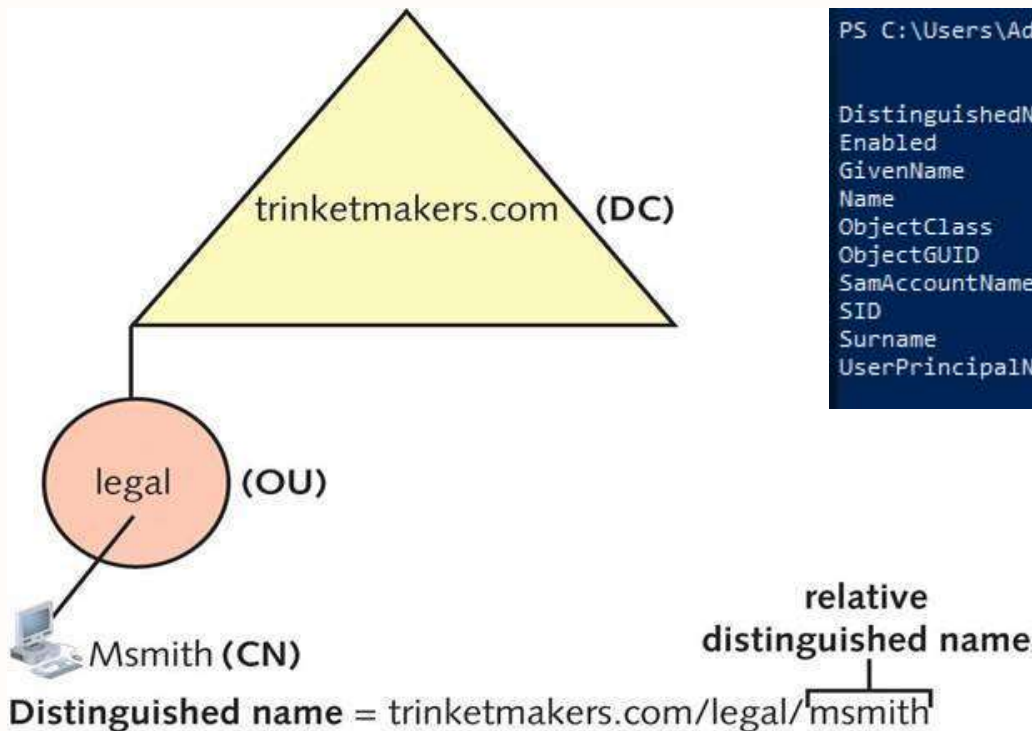


Naming Conventions

- Naming conventions based on LDAP naming conventions
- Namespace refers to collection of object names and associated places in Windows network
- Internet and Active Directory namespaces are compatible
- Each Windows network object can have multiple names
 - Distinguished name (DN)
 - Domain component (DC) name
 - Organizational unit (OU) name
 - Common name (CN): unique within a container
 - Relative distinguished name (RDN): uniquely identifies an object within a container
 - User principal name (UPN): preferred naming convention for users in e-mail, Internet services
 - Globally unique identifier (GUID): 128-bit number ensuring that no two objects have duplicate names
 - Security identifier (SID)



Naming Conventions



```
PS C:\Users\Administrator> Get-ADUser 1234567
```

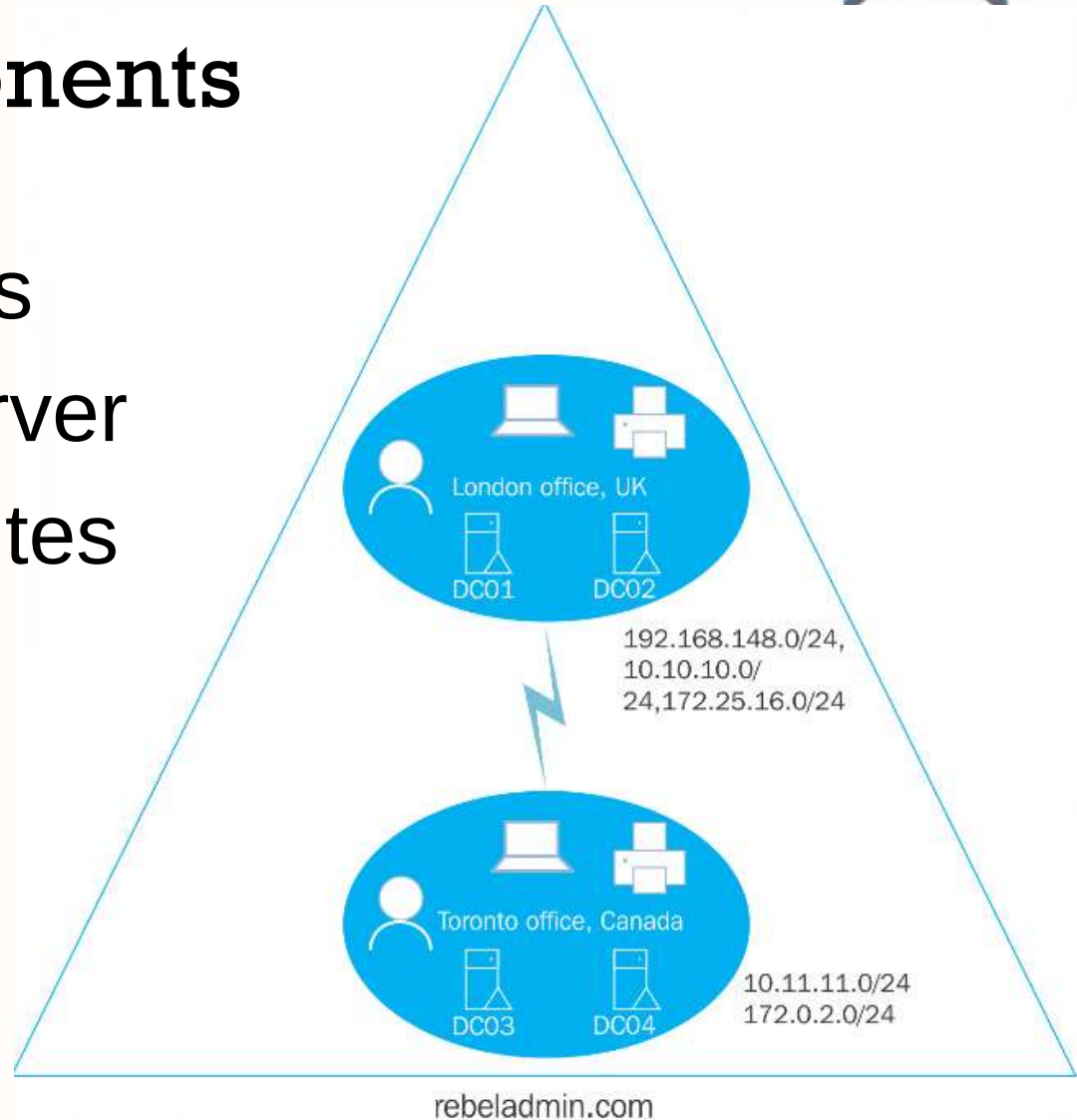
```
DistinguishedName : CN=1234567,CN=Users,DC=1234567site,DC=com
Enabled           : True
GivenName        : Consolee
Name             : 1234567
ObjectClass      : user
ObjectGUID       : 4d48ce18-495e-47d9-a671-9439d94f1a95
SamAccountName   : 1234567
SID              : S-1-5-21-977409961-1058489942-3733564020-1008
Surname          : Mbarushimana
UserPrincipalName : cmbarushimana@1234567site.com
```

UPN=msmith@trinketmakers.com



Physical Components

- Domain controllers
- Global catalog server
- Active Directory sites





Domain Controller -DC

- Central point of contact
- Storage container for all identification that happens on the network
- Active Directory Domain Services, or AD DS role
 - Turns a server into a DC
- The purpose of DC is to create the Active directory database
 - Objects: usernames, passwords, computer accounts, shared folders, printers, etc



Active Directory

- To use Active Directory, at least one server must be configured as a Domain Controller (DC).
- It is recommended that there be at least two DCs in each domain, for fault tolerance.
- Replication is the process of copying data from one computer to one or more other computers and synchronizing that data so that it is identical on all systems.
- Active Directory uses multimaster replication to copy directory information between the domain controllers in a domain.
- Each object in Active Directory has an Access Control List (ACL) that contains all access permissions associated with that object. Permissions can be either explicitly allowed or denied.



Active Directory Objects

- Objects represent network resources
- Attributes store information about an object
- In AD, there are two types of objects:
 - *Container objects* can store other objects in the Active Directory.
 - The domain itself is an example of a container object.
 - The organizational unit is also a container object.
 - *Leaf objects* cannot store other objects in Active Directory. A service account is an example of a leaf object.



Active Directory Objects

New Object - User

Create in: 1234567site.com/Users

First name: Initials:

Last name:

Full name:

User logon name: @1234567site.com

User logon name (pre-Windows 2000): 1234567SITE\

< Back Next > Cancel

New Object - Computer

Create in: 1234567site.com/Computers

Computer name:

Computer name (pre-Windows 2000):

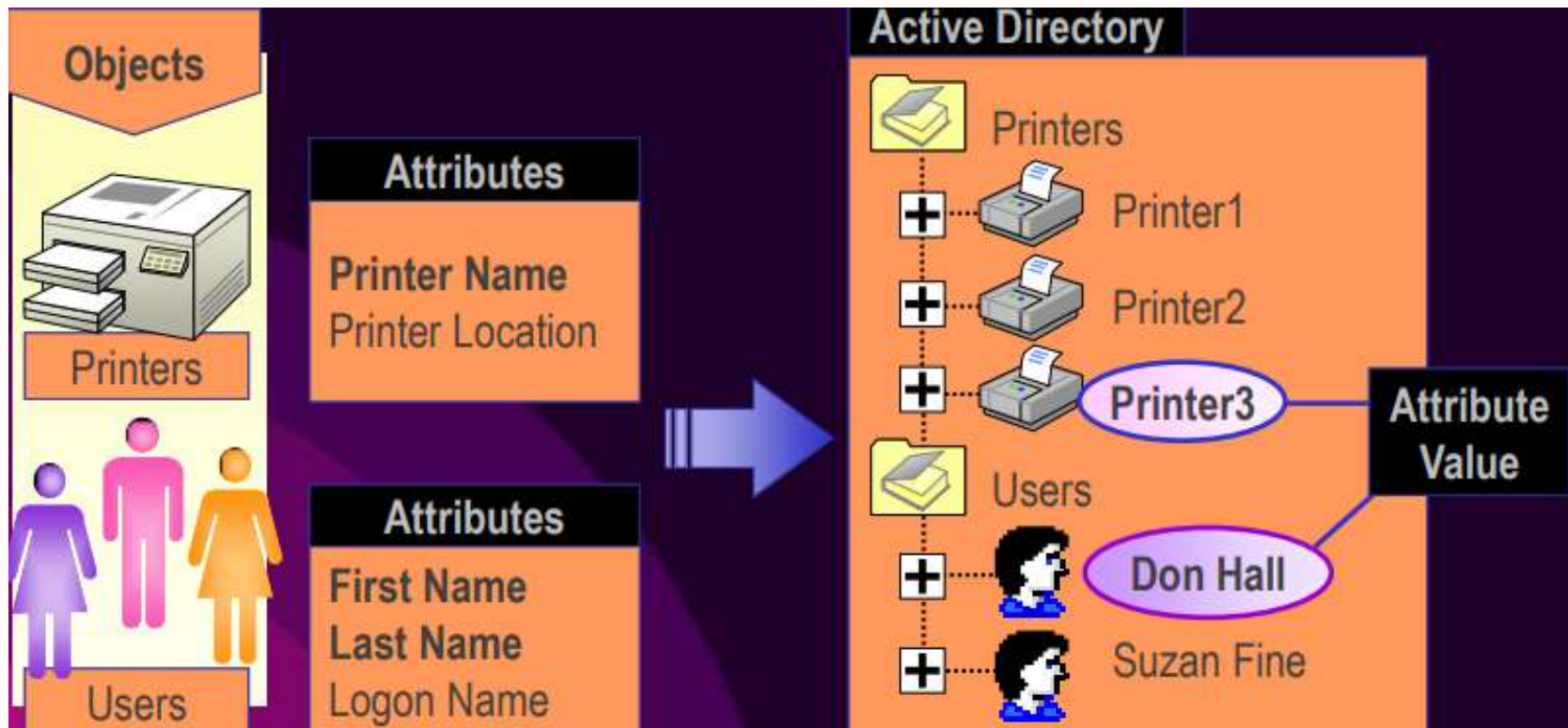
The following user or group can join this computer to a domain.

User or group: Default: Domain Admins Change...

☐ Assign this computer account as a pre-Windows 2000 computer

OK Cancel Help

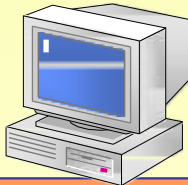
Active Directory Objects





Active Directory Schema

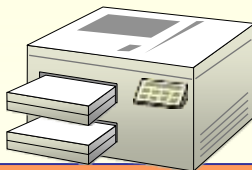
Objects Class Examples



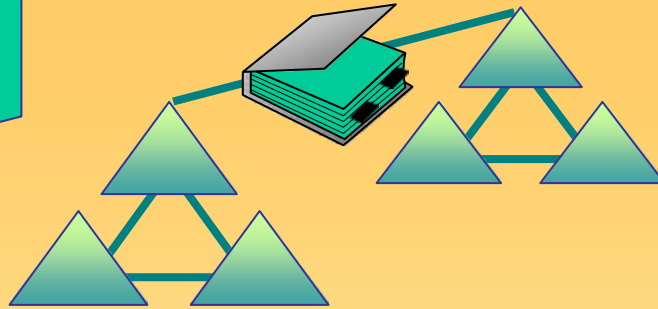
Computers



Users



Printers



Active Directory Schema Is:

- Dynamically Available
- Dynamically Updateable
- Protected by DACLs

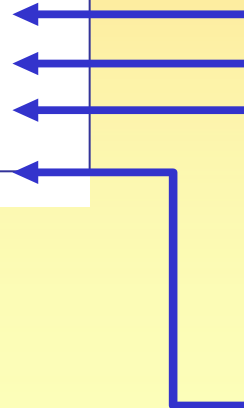
Attribute Examples

Attributes of Users Might Contain:

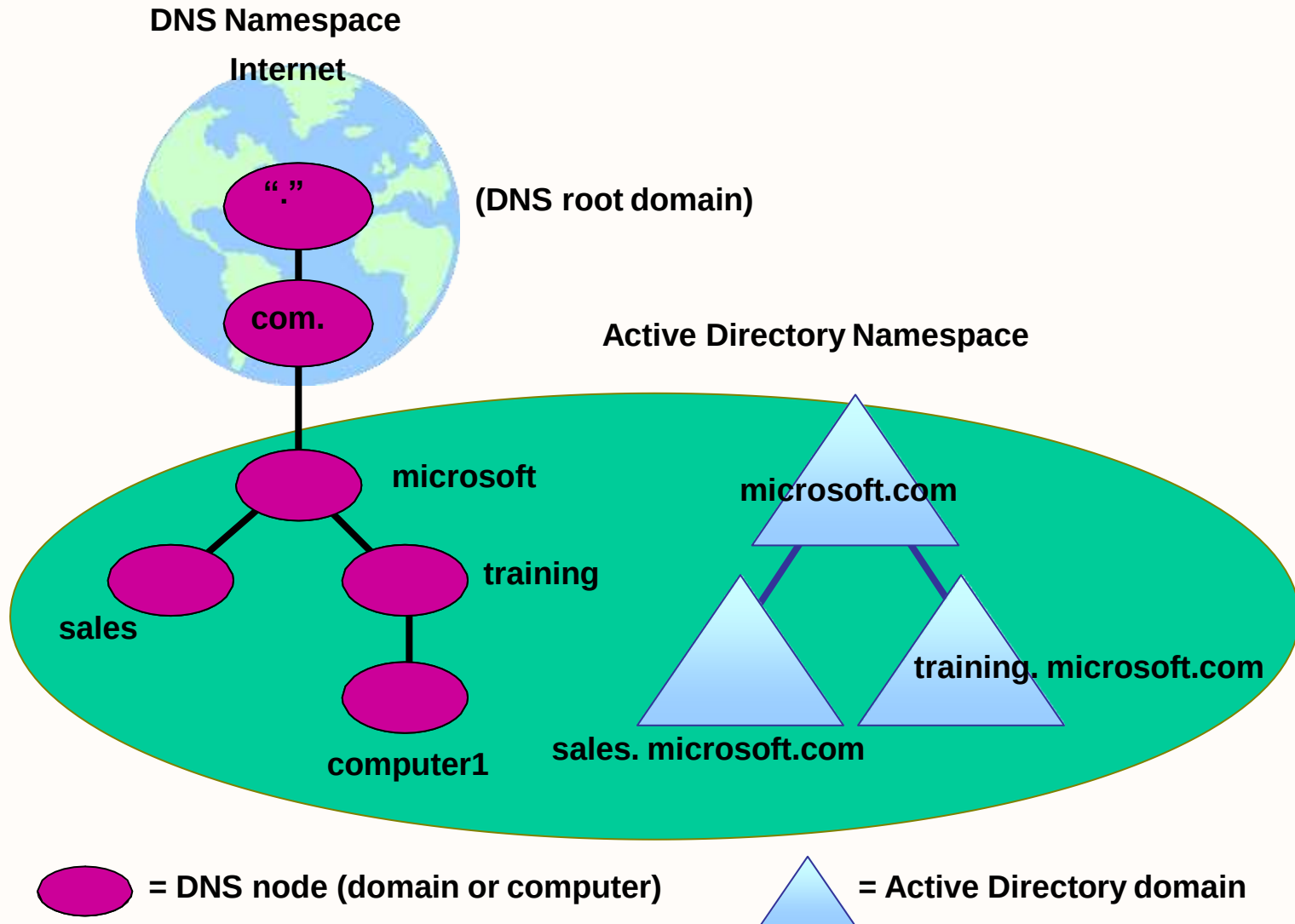
accountExpires
department
distinguishedName
middleName

List of Attributes

accountExpires
department
distinguishedName
directReports
dNSHostName
operatingSystem
repsFrom
repsTo
middleName
...



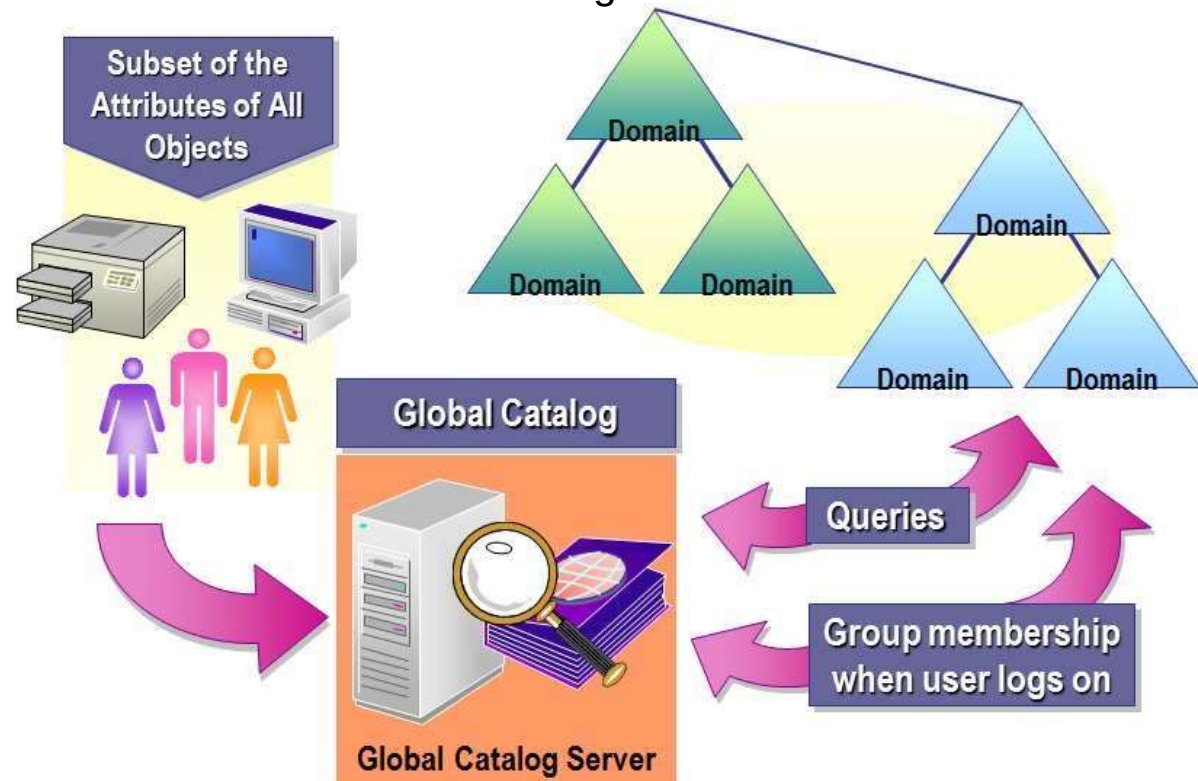
DNS and AD Namespaces





Global Catalog

- The global catalog is a distributed data repository that contains a searchable, partial representation of every object in every domain in a multi-domain Active Directory Domain Services forest.
- The global catalog is stored on domain controllers that have been designated as global catalog servers and is distributed through multi-master replication.
- Searches that are directed to the global catalog are faster because they do not involve referrals to different domain controllers





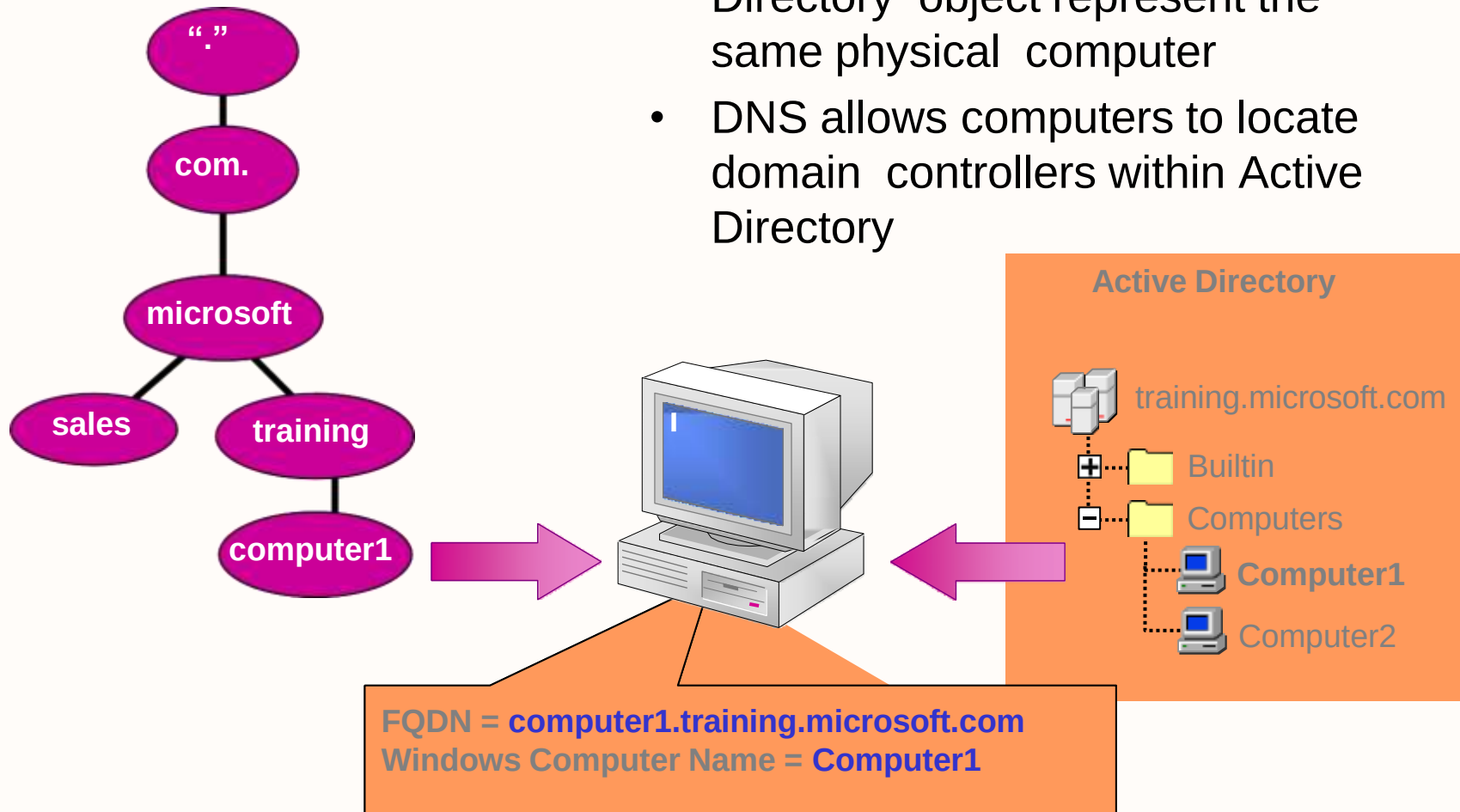
Role of DNS in Active Directory

- Name Resolution
 - DNS translates computer names to IP addresses
 - Computers use DNS to locate each other on the network
- Naming Convention for Windows Domains
 - Windows uses DNS naming standards for domain names
 - DNS domains and Active Directory domains share a common hierarchical naming structure
- Locating the Physical Components of Active Directory
 - DNS identifies domain controllers by the services they provide
 - Computers use DNS to locate domain controllers and global catalog servers



DNS Host Names and Windows Computer Names

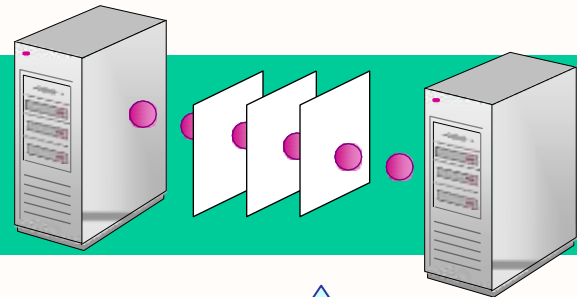
- DNS host record and Active Directory object represent the same physical computer
- DNS allows computers to locate domain controllers within Active Directory



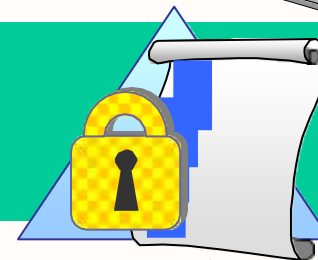


Characteristics of Multiple Domains

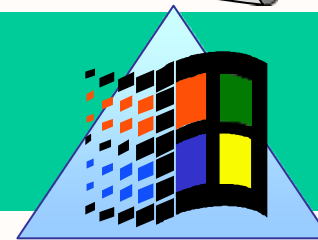
Reduce Replication Traffic



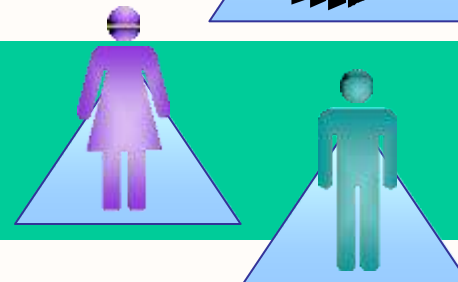
Maintain Separate and Distinct Security Policies Between Domains



Preserve the Domain Structure of Earlier Versions of Windows NT



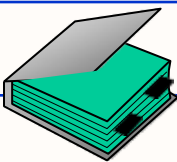
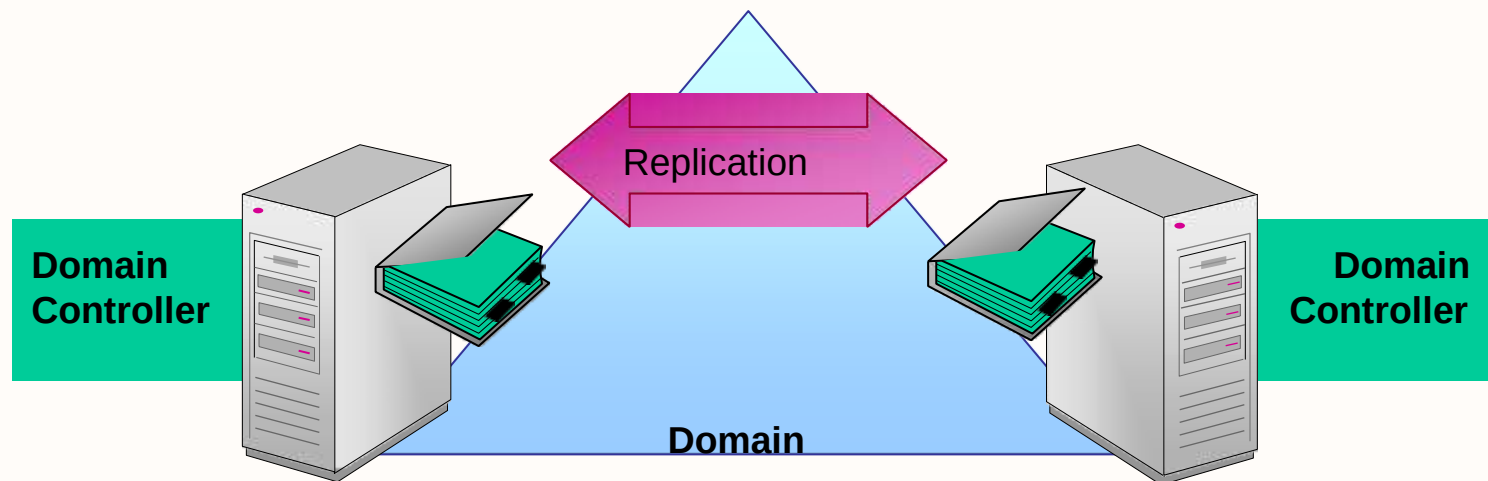
Separate Administrative Control





AD Replication

- Domain Controllers:
 - Participate in Active Directory replication
 - Perform single master operations roles in a domain



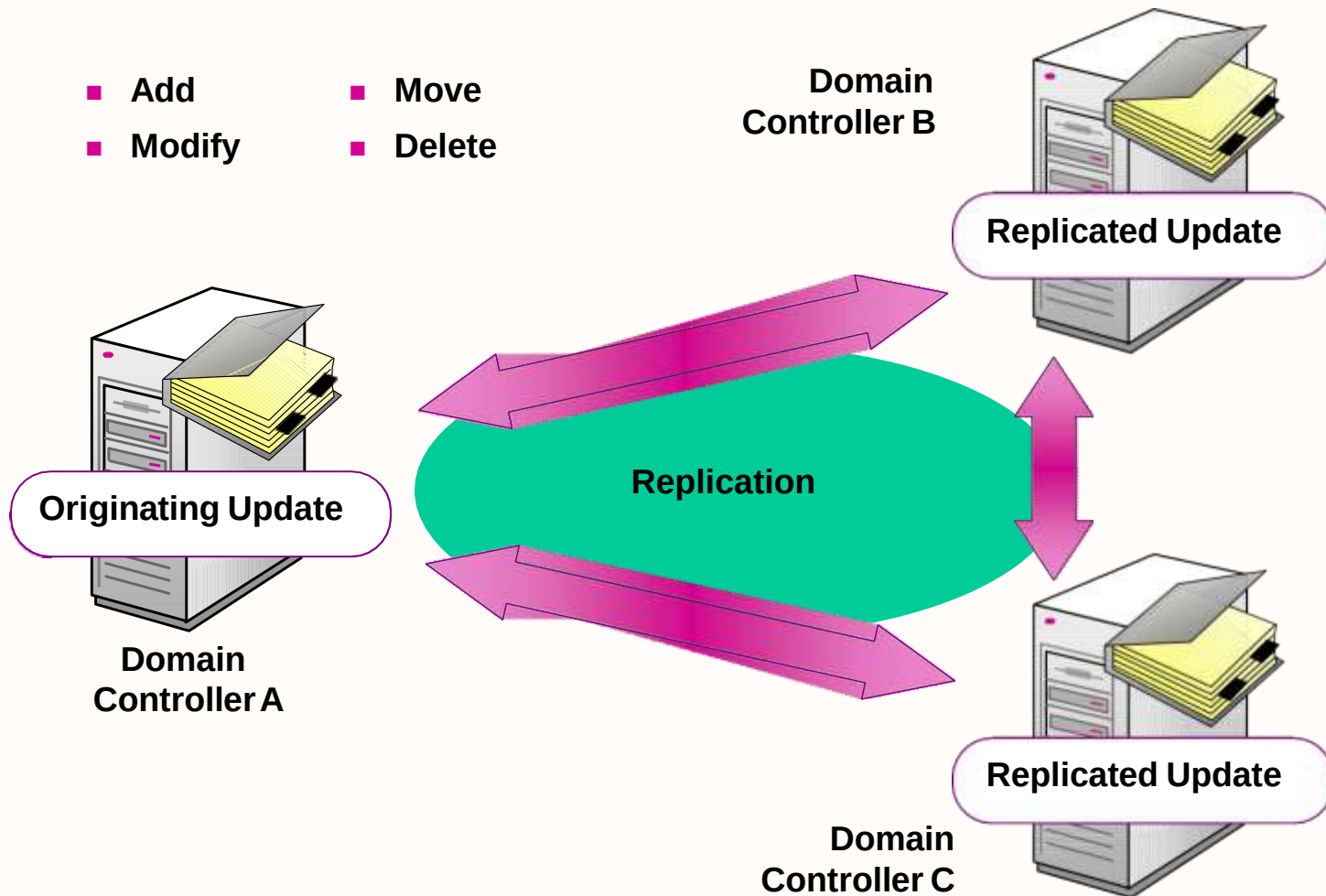
= A Writeable Copy of the Active Directory Database



How Replication Works

Active Directory Update

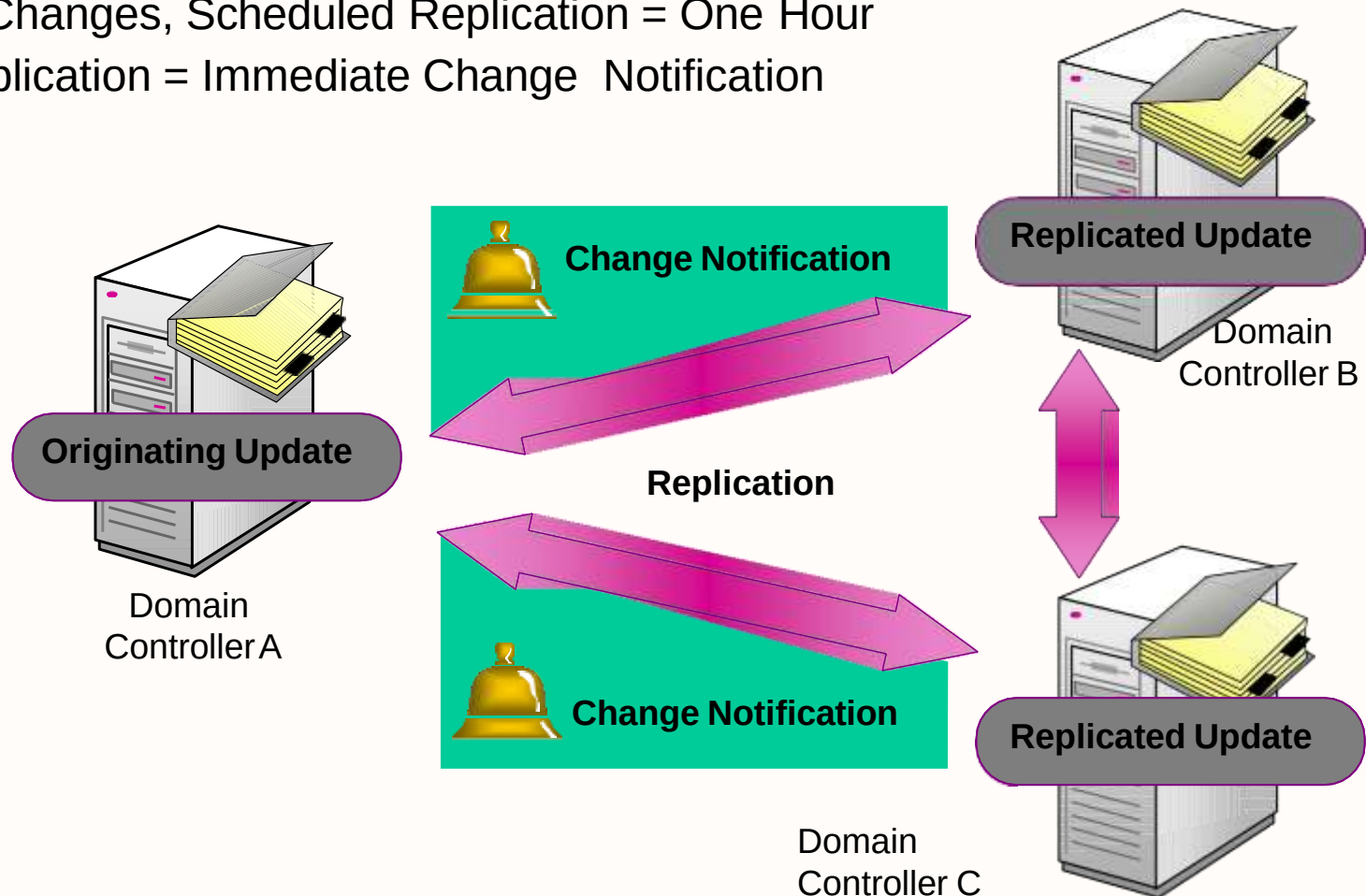
- Add
- Move
- Modify
- Delete





Replication Latency

- Default Replication Latency (Change Notification) = 5 minutes
- When No Changes, Scheduled Replication = One Hour
- Urgent Replication = Immediate Change Notification

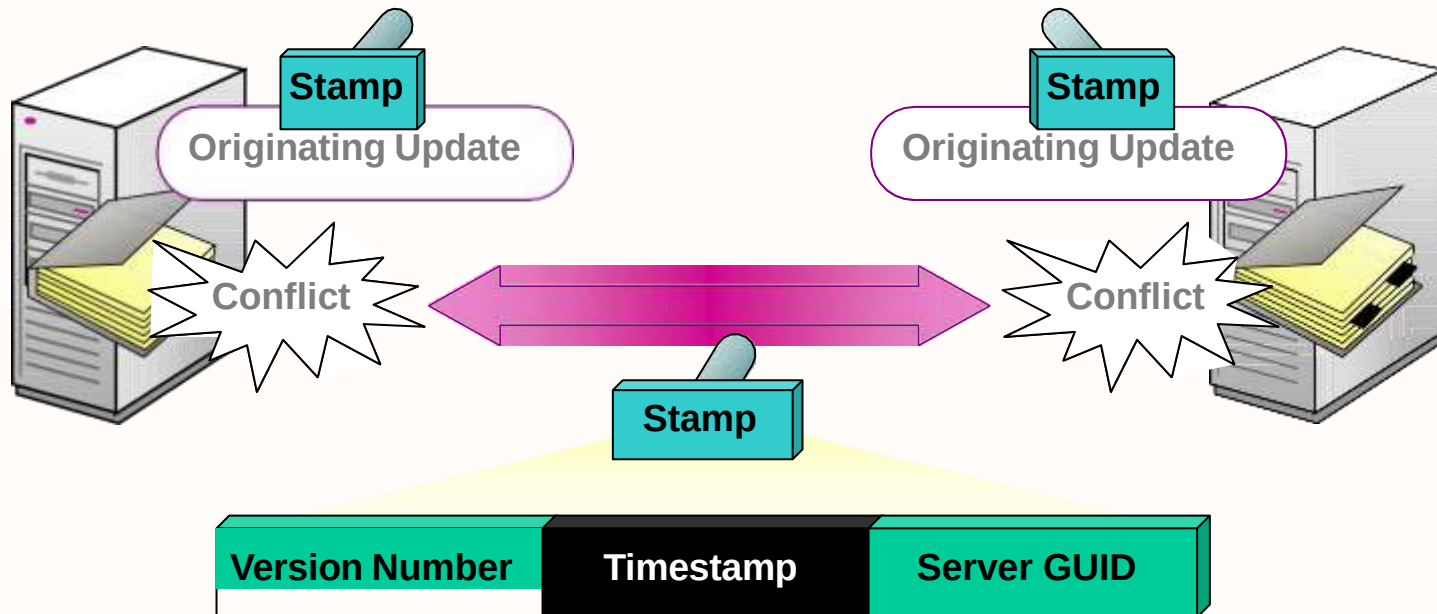




Resolving Replication Conflicts

Domain Controller A

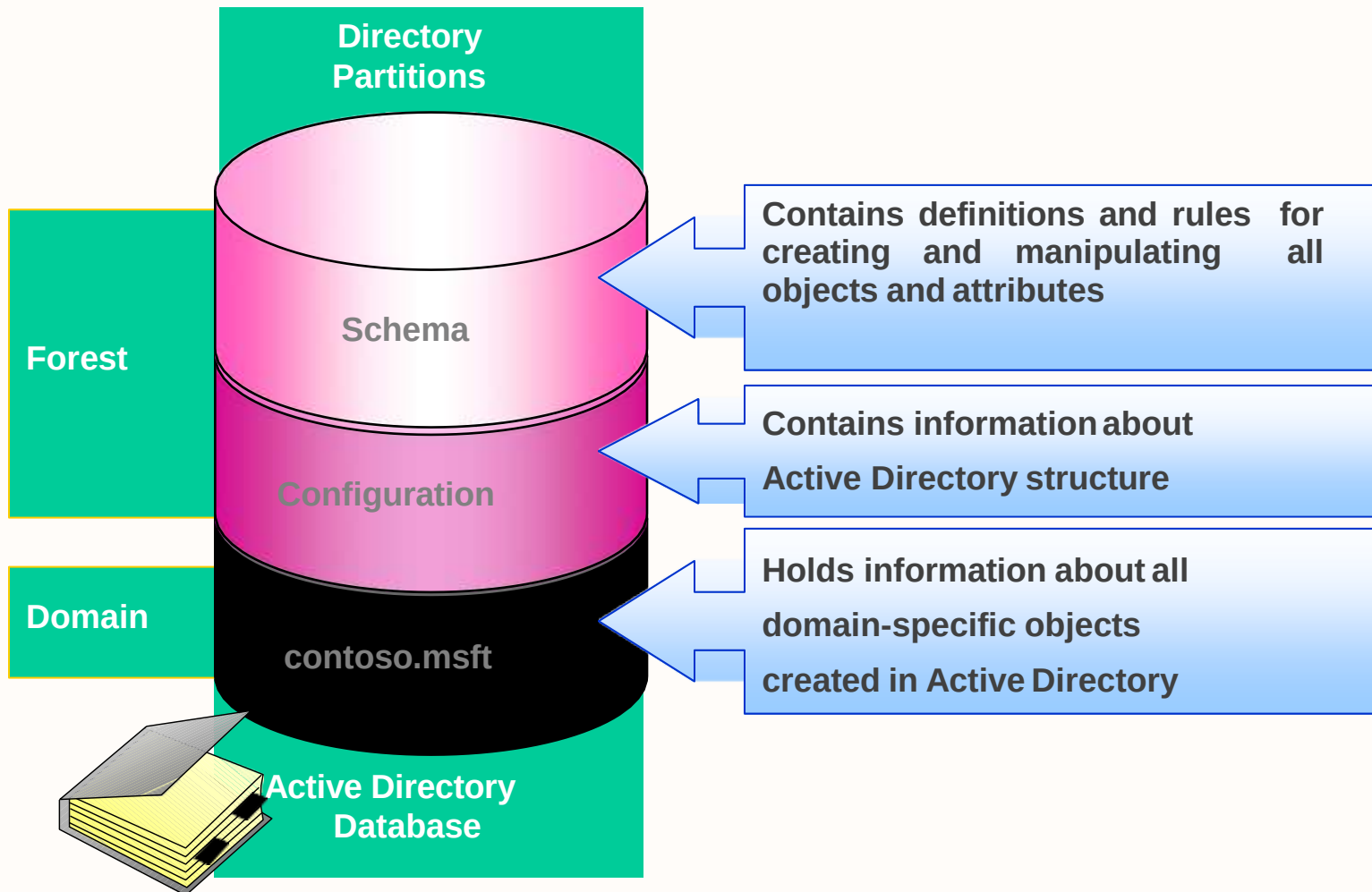
Domain Controller B



- Conflicts Can Be Due to:
 - Attribute Value
 - Adding/Moving Under a Deleted Container Object or the Deletion of a Container Object
 - Sibling Name

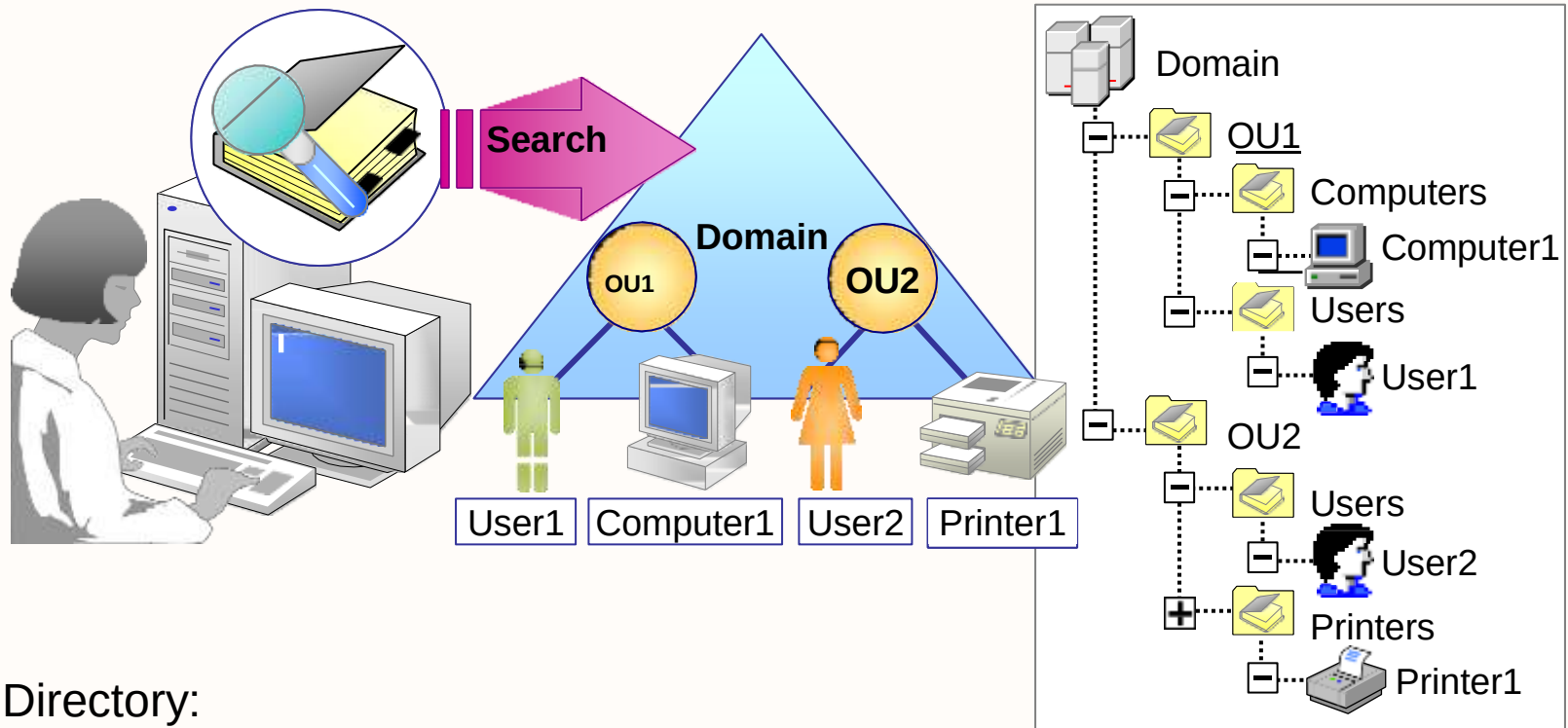


Directory Partitions





Using AD for Centralized Management

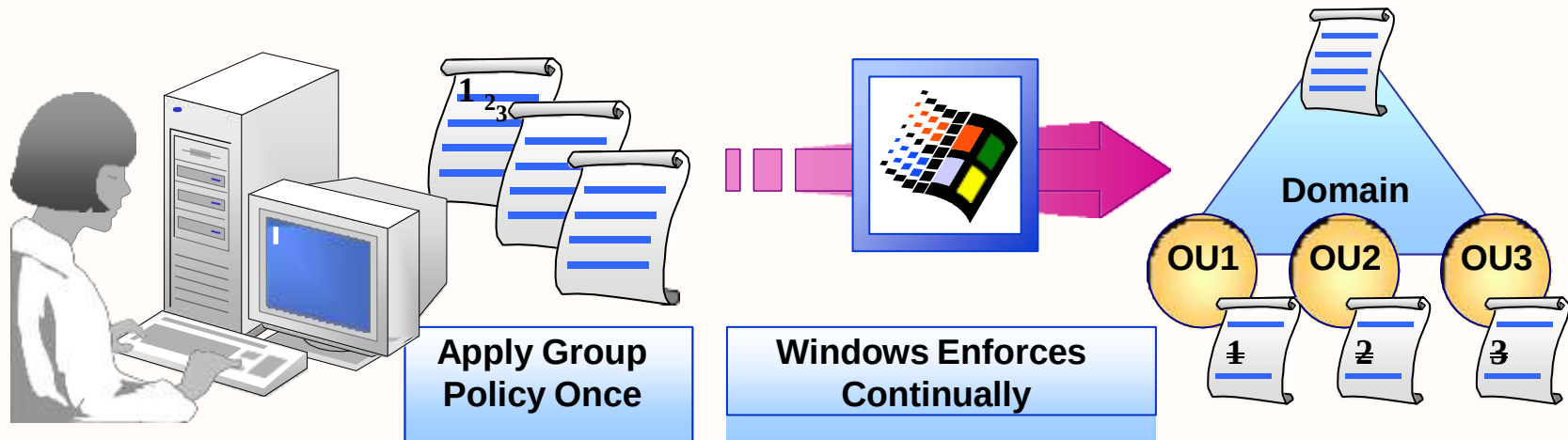


Active Directory:

- Enables a single administrator to centrally manage resources
- Allows administrators to easily locate information
- Allows administrators to group objects into OUs
- Uses Group Policy to specify policy-based settings



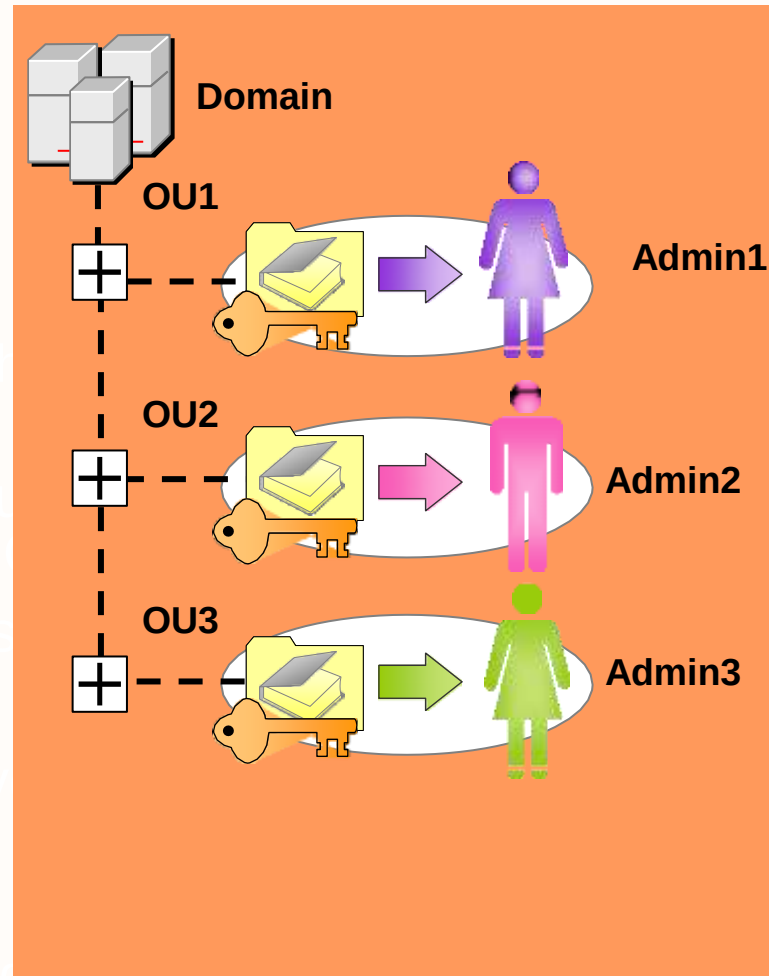
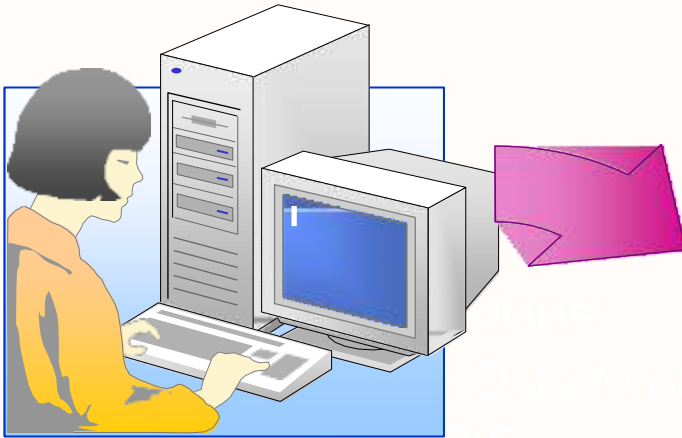
Managing the User Environment



- Use Group Policy to:
 - Control and lock down what users can do
 - Centrally manage software installation, repairs, updates, and removal
 - Configure user data to follow users whether they are online or offline



Delegating Administrative Control





Azure AD

- Moving to Azure
 - Lift and shift: Existing servers and applications can shift to the cloud using methods such as replication.
 - Build in the cloud: Organizations can directly start using pre-built Azure services and applications, and migrate existing data using methods such as replication, backup, and restore.
- Extend an existing on-premises AD environment to the cloud
 - **Site-to-site VPN or ExpressRoute:** extend your infrastructure boundary to the cloud.
 - **An additional domain controller in Azure:** set up additional domain controllers in Azure and replicate them from an on-premises Active Directory. Then, Azure will treat it as another Active Directory site.
 - **A brand new domain controller:** If it is a cloud-only environment, you can also deploy a brand new domain controller in an Azure virtual machine, and use this.



Questions?